Code No. 3528/S

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) V - Semester (Supplementary) Examinations Jul/Aug-2024

ARTIFICIAL INTELLIGENCE

Time : 3 hours **Max. Marks : 70**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each carries **14** Marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	List any two challenges faced in the field of Artificial Intelligence.	2	1	1
b)	Define a rational agent.	2	1	1
c)	What is a search tree.	2	1	1
d)	State a heuristic function.	2	2	1
e)	Define an appropriate language and formalize the following sentence using first order logic formula: "All students are smart."	2	3	2
f)	Two coins are tossed. Find the conditional probability that two heads will occur given that at least one occurs.	2	3	2
g)	List any two challenges faced in chatbot development.	2	1	3
2. a)	Demonstrate four categories of Intelligent Systems.	7	2	1
b)	Explain any one approach of solving Tic-Tac-Toe game problem.	7	2	2
3. a)	Illustrate Alpha-Beta pruning with an example.	7	2	2
b)	Consider the 3-puzzle problem with 2 x 2 board and there are three tiles, numbered 1, 2, and 3, and one blank. There are four operators, which move the blank up, down, left, and right. The start and goal states are given below. Show how the path to the goal can be found using breath first search.	7	4	2

Start

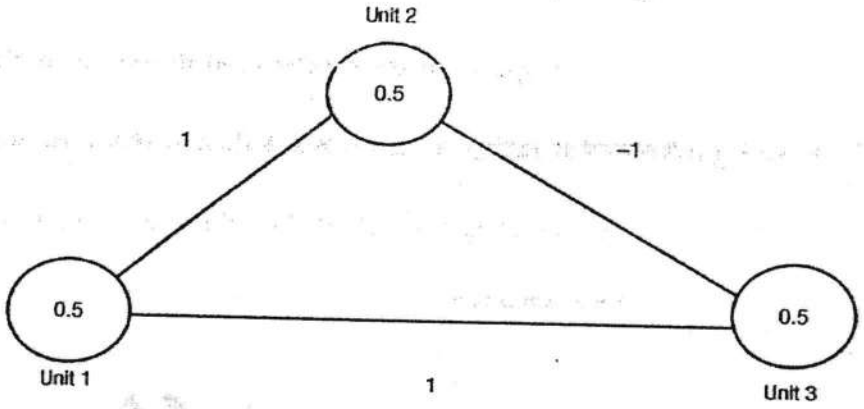
2	3
1	

Goal

1	2
	3

(P.T.O.)

4. a)	Construct knowledge representation using frames.	7	3	1
b)	Prove the following in axiomatic system: $\{A \rightarrow B, A\} \vdash (C \rightarrow B)$	7	5	2
5. a)	Let A, B, C, D be Boolean random variables. Given that: A and B are (absolutely) independent. C is independent of B given A. D is independent of C given A and B. $P(A=T) = 0.3$ $P(B=T) = 0.6$ $P(C=T\|A=T) = 0.8$, $P(C=T\|A=F) = 0.4$ $P(D=T\|A=T,B=T) = 0.7$ $P(D=T\|A=T,B=F) = 0.8$ $P(D=T\|A=F,B=T) = 0.1$ $P(D=T\|A=F,B=F) = 0.2$ Compute the following quantities: i) $P(D=F, C=T)$ ii) $P(A=T\|C=T)$	7	4	2
b)	Differentiate between expert system and traditional systems.	7	4	3
6. a)	Consider the Hopfield network drawn in the figure. Calculate the energy for each state and draw the state transition diagram for this network. Identify the stable states (if any).	7	5	2



b)	Explain recurrent networks with an example.	7	2	2
7. a)	What are local search algorithms? Explain hill climbing search.	7	1	1
b)	Demonstrate an architecture of an expert system.	7	2	1

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